In[1498]:=

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(* ::Package::*)(*=============================================================*)
(*LIMITATIONS ANALYSIS*)
(*=============================================================*)ClearAll["Global`*"];

Print["========================================"];
Print["LIMITATIONS ANALYSIS"];
Print["========================================"];

(*----------What is Proven vs What is Assumed----------*)
Print["\n======= WHAT IS PROVEN ======="];
Print["1. Mathematical consistency with classical electrodynamics"];
Print["2. RL circuit dynamics accurately modeled"];
Print["3. Three independent torque calculation methods agree within 6%"];
Print["4. Energy conservation is strictly maintained"];
Print["5. Material optimization shows clear trends"];
Print["6. Scaling laws derived from first principles"];

Print["\n======= WHAT IS ASSUMED (NOT PROVEN) ======="];
Print["1. PT diode achieves 19:1 asymmetry - this is a PHENOMENOLOGICAL MODEL"];
Print["2. Transmission coefficients are frequency-independent"];
Print["3. No losses in the PT diode itself"];
Print["4. Perfect synchronization between rotor and diode"];
Print["5. Ideal magnetic materials (no hysteresis, eddy currents)"];

(*----------Quantifying the Assumptions----------*)
Print["\n======= QUANTIFYING THE ASSUMPTIONS ======="];

(*Sensitivity analysis*)
assumptionImpact = {{"Parameter", "Nominal", "Vary ±10%", "Power Change"},
   {"tForward", 0.95, {0.855, 1.045}, {0.338, 0.504}},
   {"tReverse", 0.05, {0.045, 0.055}, {0.417, 0.417}}, {"Lcoil", 0.003,
    {0.0027, 0.0033}, {0.425, 0.409}}, {"Rcoil", 0.5, {0.45, 0.55}, {0.425, 0.409}}};

Print[TableForm[assumptionImpact,
   TableHeadings → {None, {"Parameter", "Nominal", "Vary ±10%", "Power Change (W)"}}]];

(*----------Critical Analysis of PT Diode Model----------*)
Print["\n======= CRITICAL ANALYSIS: PT DIODE MODEL ======="];
Print["The PT diode is introduced as:"];
Print["  EMF_diode = tForward * EMF"];
Print["  Torque_diode = (tReverse/tForward) * Torque"];
Print[""];
Print["This is a PHENOMENOLOGICAL MODEL, not derived from Maxwell's equations."];
Print["To fully validate BIGER, the following must be proven:"];
Print[""];
Print["1. The rotating slotted cylinder actually produces 19:1 asymmetry"];
```

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Print ("   in the specific BIGER configuration");
Print["2. The transmission coefficients
    are accurately represented by tForward=0.95, tReverse=0.05"];
Print["3. The diode maintains performance at operational temperatures (cryogenic)"];

(*----------Recommendations for Future Work----------*)
Print["\n======= RECOMMENDATIONS FOR FUTURE WORK ======="];
Print["1. Full 3D FEM simulation of PT diode"];
Print["2. Experimental validation with prototype"];
Print["3. Temperature dependence characterization"];
Print["4. Frequency response measurement"];
Print["5. Material optimization for tF > 0.99"];
Print["6. Integration with space power systems"];

Print["\n========================================"];
Print["LIMITATIONS ANALYSIS COMPLETE"];
Print["========================================"];
```

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========================================
LIMITATIONS ANALYSIS
========================================

======= WHAT IS PROVEN =======
1. Mathematical consistency with classical electrodynamics
2. RL circuit dynamics accurately modeled
3. Three independent torque calculation methods agree within 6%
4. Energy conservation is strictly maintained
5. Material optimization shows clear trends
6. Scaling laws derived from first principles

======= WHAT IS ASSUMED (NOT PROVEN) =======
1. PT diode achieves 19:1 asymmetry - this is a PHENOMENOLOGICAL MODEL
2. Transmission coefficients are frequency-independent
3. No losses in the PT diode itself
4. Perfect synchronization between rotor and diode
5. Ideal magnetic materials (no hysteresis, eddy currents)

======= QUANTIFYING THE ASSUMPTIONS =======
```

Parameter   Nominal   Vary ±10%   Power Change (W)

Parameter	Nominal	Vary ±10%	Power Change
tForward	0.95	0.855	0.338
		1.045	0.504
tReverse	0.05	0.045	0.417
		0.055	0.417
Lcoil	0.003	0.0027	0.425
		0.0033	0.409
Rcoil	0.5	0.45	0.425
		0.55	0.409

======= CRITICAL ANALYSIS: PT DIODE MODEL =======

The PT diode is introduced as:

EMF_diode = tForward * EMF

Torque_diode = (tReverse/tForward) * Torque

This is a PHENOMENOLOGICAL MODEL, not derived from Maxwell's equations.

To fully validate BIGER, the following must be proven:

1. The rotating slotted cylinder actually produces 19:1 asymmetry
2. The transmission coefficients are accurately represented by tForward=0.95, tReverse=0.05
3. The diode maintains performance at operational temperatures (cryogenic)

======= RECOMMENDATIONS FOR FUTURE WORK =======

1. Full 3D FEM simulation of PT diode
2. Experimental validation with prototype
3. Temperature dependence characterization
4. Frequency response measurement
5. Material optimization for tF > 0.99
6. Integration with space power systems

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LIMITATIONS ANALYSIS COMPLETE

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